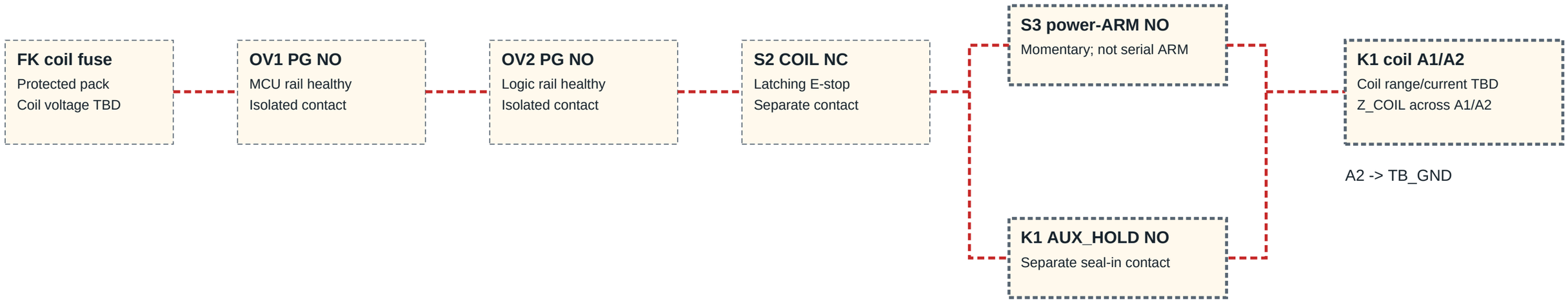
Independent emergency cutoff and manual power-arm

Functional contact roles only; actual contactor/supervisor/contact-block parts and terminal numbers must be selected and verified.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

1. Coil seal-in: independent of ESP32; S3 must be released for normal no-auto-restart behavior.

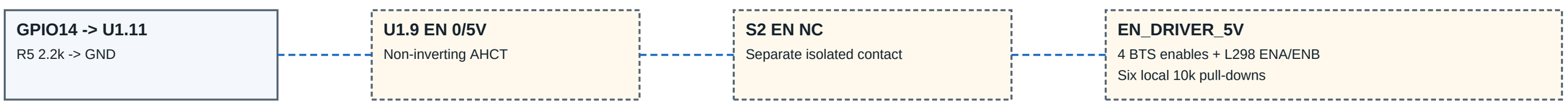


K1 MAIN_NO powers FL/FR/FA only. Stop opens coil + EN; K1 drops its seal-in and isolated feedback. Release alone does not restore power with S3 released. Held S3 / welded contacts are not covered by this functional single-channel circuit.

2. Firmware healthy input: HIGH only when E-stop released AND K1 closed.

K1-open boot intentionally latches ESTOP. Later: EN measured LOW -> deliberate power-ARM -> GPIO27 HIGH -> RESET -> ARM.

3. Shared enable: physical S2 contact opens every driver EN, independently of software.



Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.